dl.py - C:\Users\pings\Desktop\dl.py (3.14.2)

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```
#Exp No:13 Experiment to handle missing values and inappropriate data in a given data set
#Description: Understand the procedure to handle missing and inappropriate data

import numpy as np
import pandas as pd
pd.set_option('display.width', 200)
pd.set_option('display.max_columns', None)
df=pd.read_csv("Hotel_Dataset.csv")
print(df)
print(df.duplicated())
df.info()
df.drop_duplicates(inplace=True)
print(df)
print(len(df))
index=np.array(list(range(0,len(df))))
df.set_index(index,inplace=True)
print(index)
print(df)
df.drop(['Age_Group.1'],axis=1,inplace=True)
print(df)
df.loc[df.CustomerID<0,'CustomerID']=np.nan
df.loc[df.Bill<0,'Bill']=np.nan
df.loc[df.EstimatedSalary<0,'EstimatedSalary']=np.nan
print(df)
df.loc[(df['NoOfPax']<1) | (df['NoOfPax']>20),'NoOfPax']=np.nan
print(df)
print(repr(np.array(df.Age_Group.unique(),dtype=object)))
print(repr(np.array(df.Hotel.unique(),dtype=object)))
df['Hotel']=df.Hotel.replace(['Ibys'],'Ibis')
print(df.FoodPreference.unique)
df['FoodPreference']=df.FoodPreference.replace(['Vegetarian','veg'],'Veg')
df['FoodPreference']=df.FoodPreference.replace(['non-Veg'],'Non-Veg')
df['EstimatedSalary']=df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()))
df['NoOfPax']=df.NoOfPax.fillna(round(df.NoOfPax.median()))
df['Rating(1-5)']=df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()))
df['Bill']=df.Bill.fillna(round(df.Bill.mean()))
print(df)
```

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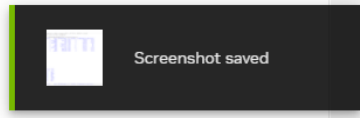
```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:\Users\pings\Desktop\dl.py ==========
    CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference  Bill  NoOfPax  EstimatedSalary Age_Group.1
0            1     20-25            4       Ibis            veg  1300        2            40000       20-25
1            2     30-35            5  LemonTree        Non-Veg  2000        3            59000       30-35
2            3     25-30            6     RedFox            Veg  1322        2            30000       25-30
3            4     20-25           -1  LemonTree            Veg  1234        2           120000       20-25
4            5       35+            3       Ibis     Vegetarian   989        2            45000         35+
5            6       35+            3       Ibys        Non-Veg  1909        2           122220         35+
6            7       35+            4     RedFox     Vegetarian  1000       -1            21122         35+
7            8     20-25            7  LemonTree            Veg  2999      -10           345673       20-25
8            9     25-30            2       Ibis        Non-Veg  3456        3           -99999       25-30
9            9     25-30            2       Ibis        Non-Veg  3456        3           -99999       25-30
10          10     30-35            5     RedFox        non-Veg  6755        4            87777       30-35
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9      True
10    False
dtype: bool
<class 'pandas.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       11 non-null     int64
 1   Age_Group        11 non-null     str
 2   Rating(1-5)      11 non-null     int64
 3   Hotel            11 non-null     str
 4   FoodPreference   11 non-null     str
 5   Bill             11 non-null     int64
 6   NoOfPax          11 non-null     int64
 7   EstimatedSalary  11 non-null     int64
 8   Age_Group.1      11 non-null     str
dtypes: int64(5), str(4)
memory usage: 1.1 KB
    CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference  Bill  NoOfPax  EstimatedSalary Age_Group.1
```

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```
 7   EstimatedSalary  11 non-null     int64
 8   Age_Group.1      11 non-null     str
dtypes: int64(5), str(4)
memory usage: 1.1 KB
    CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference  Bill  NoOfPax  EstimatedSalary Age_Group.1
0            1     20-25            4       Ibis            veg  1300        2            40000       20-25
1            2     30-35            5  LemonTree        Non-Veg  2000        3            59000       30-35
2            3     25-30            6    RedFox            Veg  1322        2            30000       25-30
3            4     20-25           -1  LemonTree            Veg  1234        2           120000       20-25
4            5       35+            3       Ibis     Vegetarian   989        2            45000         35+
5            6       35+            3       Ibys        Non-Veg  1909        2           122220         35+
6            7       35+            4    RedFox     Vegetarian  1000       -1            21122         35+
7            8     20-25            7  LemonTree            Veg  2999      -10           345673       20-25
8            9     25-30            2       Ibis        Non-Veg  3456        3           -99999       25-30
10          10     30-35            5    RedFox        non-Veg  6755        4            87777       30-35
10
[0 1 2 3 4 5 6 7 8 9]
   CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference  Bill  NoOfPax  EstimatedSalary Age_Group.1
0           1     20-25            4       Ibis            veg  1300        2            40000       20-25
1           2     30-35            5  LemonTree        Non-Veg  2000        3            59000       30-35
2           3     25-30            6    RedFox            Veg  1322        2            30000       25-30
3           4     20-25           -1  LemonTree            Veg  1234        2           120000       20-25
4           5       35+            3       Ibis     Vegetarian   989        2            45000         35+
5           6       35+            3       Ibys        Non-Veg  1909        2           122220         35+
6           7       35+            4    RedFox     Vegetarian  1000       -1            21122         35+
7           8     20-25            7  LemonTree            Veg  2999      -10           345673       20-25
8           9     25-30            2       Ibis        Non-Veg  3456        3           -99999       25-30
9          10     30-35            5    RedFox        non-Veg  6755        4            87777       30-35
   CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference  Bill  NoOfPax  EstimatedSalary
0           1     20-25            4       Ibis            veg  1300        2            40000
1           2     30-35            5  LemonTree        Non-Veg  2000        3            59000
2           3     25-30            6    RedFox            Veg  1322        2            30000
3           4     20-25           -1  LemonTree            Veg  1234        2           120000
4           5       35+            3       Ibis     Vegetarian   989        2            45000
5           6       35+            3       Ibys        Non-Veg  1909        2           122220
6           7       35+            4    RedFox     Vegetarian  1000       -1            21122
7           8     20-25            7  LemonTree            Veg  2999      -10           345673
8           9     25-30            2       Ibis        Non-Veg  3456        3           -99999
9          10     30-35            5    RedFox        non-Veg  6755        4            87777
   CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference    Bill  NoOfPax  EstimatedSalary
0         1.0     20-25            4       Ibis            veg  1300.0        2          40000.0
1         2.0     30-35            5  LemonTree        Non-Veg  2000.0        3          59000.0
2         3.0     25-30            6    RedFox            Veg  1322.0        2          30000.0
3         4.0     20-25           -1  LemonTree            Veg  1234.0        2         120000.0
4         5.0       35+            3       Ibis     Vegetarian   989.0        2          45000.0
```



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```
1            2     30-35             5  LemonTree        Non-Veg  2000        3            59000
2            3     25-30             6     RedFox            Veg  1322        2            30000
3            4     20-25            -1  LemonTree            Veg  1234        2           120000
4            5       35+             3       Ibis     Vegetarian   989        2            45000
5            6       35+             3       Ibys        Non-Veg  1909        2           122220
6            7       35+             4     RedFox     Vegetarian  1000       -1            21122
7            8     20-25             7  LemonTree            Veg  2999      -10           345673
8            9     25-30             2       Ibis        Non-Veg  3456        3           -99999
9           10     30-35             5     RedFox        non-Veg  6755        4            87777
   CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference    Bill  NoOfPax  EstimatedSalary
0         1.0     20-25            4       Ibis             veg  1300.0        2          40000.0
1         2.0     30-35            5  LemonTree        Non-Veg  2000.0        3          59000.0
2         3.0     25-30            6     RedFox            Veg  1322.0        2          30000.0
3         4.0     20-25           -1  LemonTree            Veg  1234.0        2         120000.0
4         5.0       35+            3       Ibis     Vegetarian   989.0        2          45000.0
5         6.0       35+            3       Ibys        Non-Veg  1909.0        2         122220.0
6         7.0       35+            4     RedFox     Vegetarian  1000.0       -1          21122.0
7         8.0     20-25            7  LemonTree            Veg  2999.0      -10         345673.0
8         9.0     25-30            2       Ibis        Non-Veg  3456.0        3              NaN
9        10.0     30-35            5     RedFox        non-Veg  6755.0        4          87777.0
   CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference    Bill  NoOfPax  EstimatedSalary
0         1.0     20-25            4       Ibis             veg  1300.0      2.0          40000.0
1         2.0     30-35            5  LemonTree        Non-Veg  2000.0      3.0          59000.0
2         3.0     25-30            6     RedFox            Veg  1322.0      2.0          30000.0
3         4.0     20-25           -1  LemonTree            Veg  1234.0      2.0         120000.0
4         5.0       35+            3       Ibis     Vegetarian   989.0      2.0          45000.0
5         6.0       35+            3       Ibys        Non-Veg  1909.0      2.0         122220.0
6         7.0       35+            4     RedFox     Vegetarian  1000.0      NaN          21122.0
7         8.0     20-25            7  LemonTree            Veg  2999.0      NaN         345673.0
8         9.0     25-30            2       Ibis        Non-Veg  3456.0      3.0              NaN
9        10.0     30-35            5     RedFox        non-Veg  6755.0      4.0          87777.0
array(['20-25', '30-35', '25-30', '35+'], dtype=object)
array(['Ibis', 'LemonTree', 'RedFox', 'Ibys'], dtype=object)
<bound method Series.unique of 0           veg
1       Non-Veg
2           Veg
3           Veg
4    Vegetarian
5       Non-Veg
6    Vegetarian
7           Veg
8       Non-Veg
9       non-Veg
Name: FoodPreference, dtype: str>
   CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference    Bill  NoOfPax  EstimatedSalary
```

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```
1          2.0     30-35              5  LemonTree        Non-Veg  2000.0         3          59000.0
2          3.0     25-30              6     RedFox            Veg  1322.0         2          30000.0
3          4.0     20-25             -1  LemonTree            Veg  1234.0         2         120000.0
4          5.0       35+              3       Ibis     Vegetarian   989.0         2          45000.0
5          6.0       35+              3       Ibys        Non-Veg  1909.0         2         122220.0
6          7.0       35+              4     RedFox     Vegetarian  1000.0        -1          21122.0
7          8.0     20-25              7  LemonTree            Veg  2999.0       -10         345673.0
8          9.0     25-30              2       Ibis        Non-Veg  3456.0         3              NaN
9         10.0     30-35              5     RedFox        non-Veg  6755.0         4          87777.0
   CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference    Bill  NoOfPax  EstimatedSalary
0         1.0     20-25            4       Ibis             veg  1300.0      2.0          40000.0
1         2.0     30-35            5  LemonTree        Non-Veg  2000.0      3.0          59000.0
2         3.0     25-30            6     RedFox            Veg  1322.0      2.0          30000.0
3         4.0     20-25           -1  LemonTree            Veg  1234.0      2.0         120000.0
4         5.0       35+            3       Ibis     Vegetarian   989.0      2.0          45000.0
5         6.0       35+            3       Ibys        Non-Veg  1909.0      2.0         122220.0
6         7.0       35+            4     RedFox     Vegetarian  1000.0      NaN          21122.0
7         8.0     20-25            7  LemonTree            Veg  2999.0      NaN         345673.0
8         9.0     25-30            2       Ibis        Non-Veg  3456.0      3.0             NaN
9        10.0     30-35            5     RedFox        non-Veg  6755.0      4.0          87777.0
array(['20-25', '30-35', '25-30', '35+'], dtype=object)
array(['Ibis', 'LemonTree', 'RedFox', 'Ibys'], dtype=object)
<bound method Series.unique of 0           veg
1       Non-Veg
2           Veg
3           Veg
4     Vegetarian
5       Non-Veg
6     Vegetarian
7           Veg
8       Non-Veg
9       non-Veg
Name: FoodPreference, dtype: str>
   CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference    Bill  NoOfPax  EstimatedSalary
0         1.0     20-25            4       Ibis            Veg  1300.0      2.0          40000.0
1         2.0     30-35            5  LemonTree        Non-Veg  2000.0      3.0          59000.0
2         3.0     25-30            6     RedFox            Veg  1322.0      2.0          30000.0
3         4.0     20-25           -1  LemonTree            Veg  1234.0      2.0         120000.0
4         5.0       35+            3       Ibis            Veg   989.0      2.0          45000.0
5         6.0       35+            3       Ibis        Non-Veg  1909.0      2.0         122220.0
6         7.0       35+            4     RedFox            Veg  1000.0      2.0          21122.0
7         8.0     20-25            7  LemonTree            Veg  2999.0      2.0         345673.0
8         9.0     25-30            2       Ibis        Non-Veg  3456.0      3.0          96755.0
9        10.0     30-35            5     RedFox        Non-Veg  6755.0      4.0          87777.0
>>>
```

Screenshot saved